JAVA PROGRAMMING PRACTICES

Longest Substring Without Repeating Characters

Basic Information

Week	Week 6
Source file	LongestUniqueSubstring.java
Concepts	String handling, charAt(), indexOf(), length(), loops.

Objective / Aim

To implement longest substring without repeating characters using the concepts listed in the R23 Java Programming Practices syllabus.

System Requirements

JDK (Java Development Kit) and a text editor or Java IDE.

Algorithm / Logic Description

1. Read a string.
2. Build candidate substrings without repeated characters.
3. Keep and display the longest substring and its length.

UML / Program Structure

```
LongestUniqueSubstring
main() demonstrates the algorithm above.
```

Source Code

```
import java.util.Scanner;
public class LongestUniqueSubstring
{
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a string: ");
        String text = sc.nextLine(), longest = "";
        for (int i = 0; i < text.length(); i++)
        {
            String current = "";
            for (int j = i; j < text.length(); j++)
            {
                char ch = text.charAt(j);
                if (current.indexOf(ch) >= 0) break;
                current += ch;
            }
            if (current.length() > longest.length()) longest = current;
        }
        System.out.println("Longest substring: " + longest);
        System.out.println("Length: " + longest.length());
        sc.close();
    }
}
```

Compilation and Execution

```
javac LongestUniqueSubstring.java
java LongestUniqueSubstring
```

Output Screen

```
Enter a string: Longest substring: abc
Length: 3
```

Result

The program demonstrates the listed syllabus concepts. The sample output is based on a compiled execution.